

# Characterizing Complex Hydrological Extremes: A Framework Integrating Nonstationarity Assessment and Copula-Based Analysis

Achala Singh, S.M.ASCE<sup>1</sup>; Priyank J. Sharma<sup>2</sup>; and Ramesh S. V. Teegavarapu<sup>3</sup>

**Abstract:** Robust flood frequency analysis (FFA) to assess the risk of extreme floods is achievable if multiple flood characteristics, such as peak, volume, and duration, are considered. This study presents a novel framework that integrates nonstationarity (NS) detection with univariate and bivariate assessments of flood extremes using copula functions under both disjunction and conjunction scenarios. Unlike most existing studies that adopted either univariate block maxima (BM) or peaks-over-threshold (POT) approaches in isolation, this work provides a comprehensive comparison by employing both approaches within a copula-based bivariate framework across multiple catchments, addressing a key methodological gap in FFA research. The framework was applied to 31 physioclimatically diverse catchments across four major river basins in west-central India. Results reveal spatially heterogeneous trends and frequent occurrences of NS, even in the absence of statistically significant trends. The POT approach proved more effective in accounting for multiple extreme events within a season or year, yielding more-conservative and -reliable risk estimates with comparatively narrower uncertainty bounds. Dependence analysis using multiple association measures showed the strongest correlation for peak–volume (P-V) and the weakest for peak–duration (P-D) pairs. Bivariate FFA indicated that Clayton and Gumbel copulas provided the best fit across catchments and flood characteristic pairs. Pronounced differences between univariate and bivariate return levels highlight the importance of POT-based multivariate analysis for characterizing complex hydrological dynamics. Based on the results, the catchments in the Narmada basin, India, exhibited the highest flood risk in both univariate and bivariate assessments, whereas those in the Tapi basin, India, had the highest joint risk for peak and volume floods. The proposed approach provides a realistic and regionally adaptable framework for flood quantile and risk estimation. By introducing a generic methodology for comprehensive FFA, this study facilitates the development of improved early warning systems and the design of climate-resilient water infrastructure across diverse hydrological regions. **DOI:** [10.1061/JHYEFF.HEENG-6748](https://doi.org/10.1061/JHYEFF.HEENG-6748). © 2026 American Society of Civil Engineers.

**Author keywords:** Nonstationarity; Copula; Multivariate analysis; Flood frequency analysis (FFA).

## Introduction

Floods are among the most catastrophic natural disasters, and their frequency and intensity are increasing due to climate change and human interventions (Feng et al. 2020). A cornerstone of hydrologic designs is flood frequency analysis (FFA), which estimates the recurrence intervals of extreme flood events. However, the assumption of stationarity in such designs has become increasingly obsolete amid anthropogenic climate change (Milly et al. 2008). Existence of nonstationary (NS) conditions underscores the need for greater precision and reduced uncertainty in FFA for effective flood risk management (Bezák et al. 2014). Statistically, risk is defined as the occurrence probability of one or more floods exceeding the design flood over the design

life (Salas and Obeysekera 2014). In practice, flood risk refers to the potential for adverse impacts from flooding, including loss of life, damage to property and infrastructure, and disruption of economic and social activities (Maranzoni et al. 2023). Traditional FFA often uses the block maxima (BM) approach, which simplifies analysis by selecting only the highest flow value from each year or season to create an annual maximum series (AMS). FFA, using the BM approach, often ignores other significant, independent flood events within the same year or season, thereby allowing years with peak values well below flood thresholds to unduly influence the analysis (Bezák et al. 2014). The peaks-over-threshold (POT) approach, which overcomes the drawbacks of the BM approach, enables the extraction of all peak values exceeding a predefined flood threshold, thereby capturing multiple flood events within a year or season (Pan et al. 2023). The POT approach's requirement for ensuring data independence and threshold selection adds complexity, contributing to its relatively limited practical application (Pan et al. 2023). Currently, the literature lacks rigorous comparisons of BM and POT models across diverse hydroclimatic conditions.

Although FFA is widely applied, most studies focus on flood peaks, and neglect critical characteristics such as volume, duration, and timing, which are essential for a comprehensive understanding of flood dynamics (Reddy and Ganguli 2012). In particular, the assessment of stationarity in flood characteristics remains unexplored, despite its importance for understanding the shifting nature of flood risks under an evolving climate. Flood events are inherently stochastic and characterized by interdependent properties, such as

<sup>1</sup>Ph.D. Research Scholar, Hydroinformatics Lab, Dept. of Civil Engineering, Indian Institute of Technology Indore, Indore, Madhya Pradesh 453552, India.

<sup>2</sup>Assistant Professor, Hydroinformatics Lab, Dept. of Civil Engineering, Indian Institute of Technology Indore, Indore, Madhya Pradesh 453552, India (corresponding author). ORCID: <https://orcid.org/0000-0002-0188-4923>. Email: priyanksharma@iiti.ac.in

<sup>3</sup>Professor, Dept. of Civil, Environmental and Geomatics Engineering, Florida Atlantic Univ., Boca Raton, FL 33431.

Note. This manuscript was submitted on July 28, 2025; approved on February 25, 2026; published online on June 29, 2026. Discussion period open until November 29, 2026; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Hydrologic Engineering*, © ASCE, ISSN 1084-0699.

flood peak, volume, and duration, which collectively shape the flood hydrograph (Kiran and Srinivas 2022). To effectively assess the associated risks, a robust probabilistic framework must account for the complex dependencies among these variables (Zhang and Singh 2006). Copulas (Sklar 1973) offer a flexible framework for constructing multivariate distributions by linking different marginal distributions, thereby enabling the accurate representation of dependencies among flood characteristics. Archimedean copulas are widely used in hydrologic studies due to their simplicity and ability to capture both positive and negative dependence structures (Binoy et al. 2023). Although extensive research exists in FFA, no study has yet integrated univariate and bivariate estimates using both BM and POT approaches within a single, unified framework. This study addresses that gap by providing a comprehensive comparison of these methods across diverse catchments, evaluating their strengths in representing complex flood characteristics. The analysis also incorporates stationarity assessments to account for evolving climatic and anthropogenic risks. By applying this methodology to major west-flowing river basins in India, the research offers generalizable insights that enhance design reliability and infrastructure resilience at a regional scale.

The west-flowing rivers of India, particularly the Narmada, Tapi, Mahi, and Sabarmati, which are highly susceptible to frequent, synchronous, and severe flooding due to the region's diverse climatic and physiographic characteristics (Das and Scaringi 2021; Deshpande et al. 2016), make them ideal for assessing the proposed framework, which is inherently transferable to other flood-prone regions worldwide. This study answers the following research questions:

1. Are visible patterns of nonstationarity in flood characteristics (flood peak, volume, and duration), derived from AMS and POT approaches, evident at the river basin scale?
2. How practical are the AMS and POT approaches in estimating flood quantiles across physioclimatically diverse catchments?
3. How do univariate and bivariate flood frequency analyses characterize AMS- and POT-derived flood extremes?

## Study Domain and Data Sources

The application of the proposed methodology was demonstrated using streamflow data from India's principal west-flowing river systems, namely, the Sabarmati, Mahi, Narmada, and Tapi Rivers, which drain into the Arabian Sea in the Gulf of Khambhat, Gujarat (Singh et al. 2024). The Narmada, Tapi, and Mahi Rivers originate from Peninsular India (Vindhya and Satpura Ranges), whereas the Sabarmati River originates from the Aravalli Ranges. The combined area of all these river basins is approximately 220,457 km<sup>2</sup>, located near the tropics between latitudes 20°9' and 24°55' N and longitudes 72°34' and 78°18' E. The Narmada River and the Tapi River are the two most prominent west-flowing rivers in the peninsular region, with similar narrow, elongated shapes and comparable topographic features (Singh et al. 2024). Conversely, the Mahi and Sabarmati basins are relatively broader but exhibit similar terrain (Fig. 1). Historical records indicate recurrent major floods in the Tapi (1970, 1994, 1998, 2006, 2013), Narmada (1970, 1973, 1984, 1990, 1994, 2006, 2013, 2023), Sabarmati (1941, 1950, 1973, 1992, 2006) and Mahi (1973, 1976, 1990, 1994, 2006) basins (Timbadiya et al. 2014; Sridhar et al. 2014; Mangukiya et al. 2022). Daily observed discharge data from 31 stream gauging stations in the study area were obtained from India-WRIS and the Central Water Commission. Table S1 presents a detailed summary of the datasets utilized in this study, including their sources.

## Methodological Framework

### Statistical Testing Procedures

#### Extraction of Flood Characteristics

This study employed two approaches for flood frequency analysis: block maxima, and peaks-over-threshold. The BM approach selects the maximum annual discharge to yield an AMS, and the POT approach accounts for multiple independent peaks above a specified threshold that the annual block constraint otherwise would omit. With more observations, parameter estimation for flood quantiles improves, making POT estimates more accurate (Caires 2016). Despite its advantages, the POT approach poses challenges in determining an appropriate threshold and ensuring independence between events. This study utilized the Select Flood Events and Identify Flood Characteristics (SFE\_IFC) toolbox in MATLAB R2023a, which employs an automated procedure for threshold selection and identification of flood events and their characteristics (Zhang et al. 2021). The automatic selection procedure to address the two fundamental issues with the POT approach was as follows:

1. Independence criteria: The declustering method (Durocher et al. 2018) was employed to ensure the independence of the peak discharge events selected above the threshold by meeting the following conditions:

$$t > 5 + \ln\left(\frac{A}{1.609^2}\right), \quad Q_{\min} < 0.75 \times \min(Q_1, Q_2) \quad (1)$$

where  $t$  = time interval between consecutive flood peaks (days);  $Q_{\min}$  = lowest intermediate flow recorded (m<sup>3</sup>/s);  $A$  = catchment area (km<sup>2</sup>); and  $Q_1$  and  $Q_2$  = magnitudes of two successive flood peaks (m<sup>3</sup>/s).

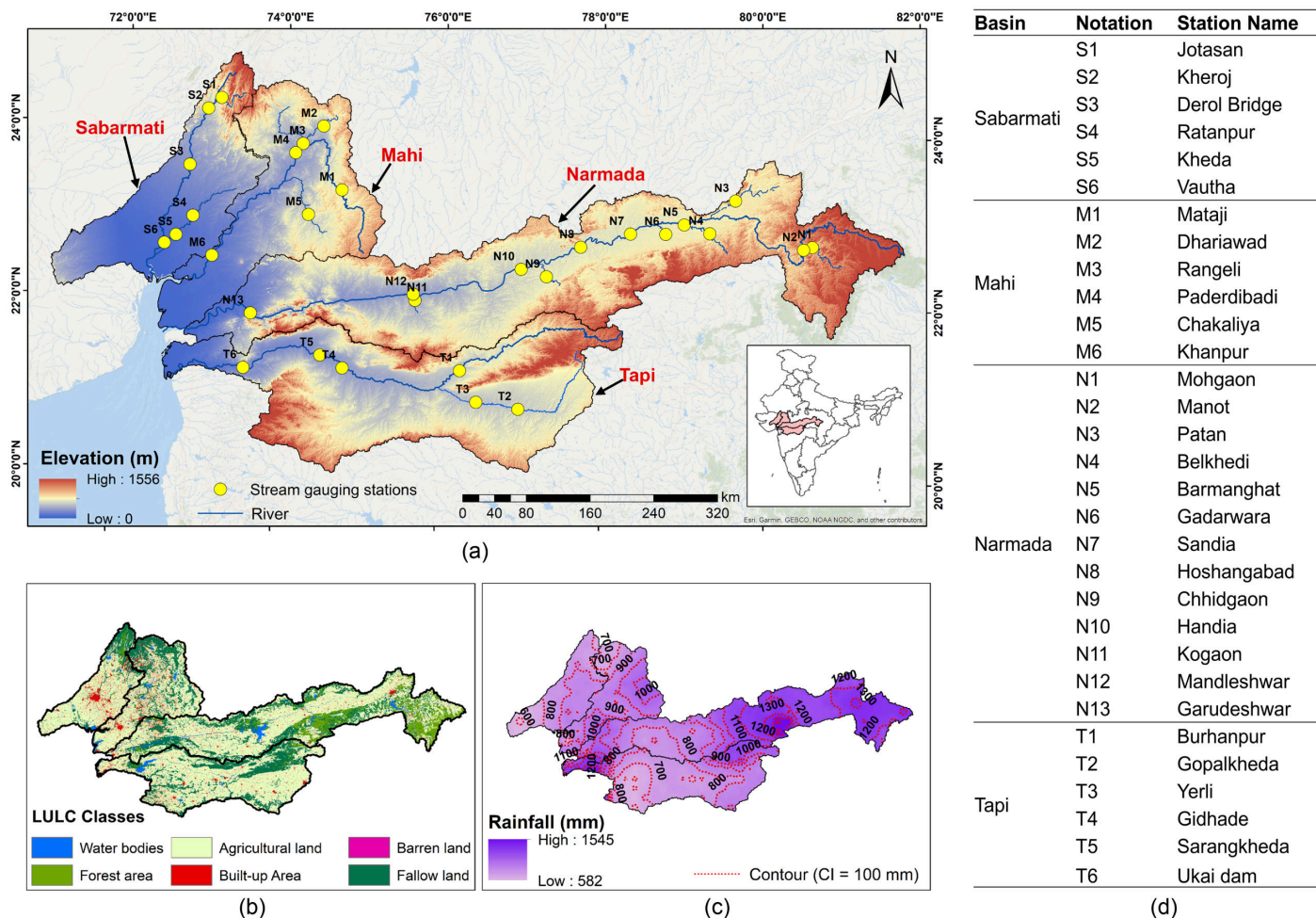
2. Threshold selection: An automatic threshold selection method for the POT analysis is based on the goodness-of-fit (GOF) approach developed by Solari et al. (2017). This test employs the generalized Pareto distribution (GPD) as the reference model, commonly used for modeling POT data. More details were provided by Zhang et al. (2021).

Following optimal threshold selection, the resulting flood peak series is evaluated for independence using Kendall's tau correlation and conformity to a Poisson distribution using the dispersion index (Cunnane 1979). In addition, a Wald-Wolfowitz runs test (Wald and Wolfowitz 1943) is performed to ensure the independence of the peak series derived from the POT approach. After confirming the independence of flood events, their characteristics are derived. The start and end points of each flood event are extracted from the hydrograph to assess event duration.

#### Stationarity Assessment

Stationarity refers to the condition in which the statistical characteristics of hydrologic variables, such as mean, variance, and distribution, remain constant over time. The absence of a unit root, trend, or change point does not necessarily indicate stationarity, because stationarity implies that a time series' statistical properties remain constant over time (Singh et al. 2024). Thus, assessing nonstationarity in flood characteristics is necessary to understand flood dynamics. Flood characteristics derived from both the AMS and POT methods are assessed for stationarity using the moving window trend analysis (Text S1) and the nonoverlapping block-stratified random sampling (NBRS) method introduced by Teegavarapu and Sharma (2021). This approach overcomes the limitations of conventional trend and change-point tests by dividing the time series into nonoverlapping blocks of equal or varying lengths, initiated





**Fig. 1.** (a) Location of stream gauging stations and elevation variability; (b) land-use land-cover information; (c) annual rainfall variability; and (d) list of station names. Notably, ~ 77.5% of the land is within the 100–500-m elevation range. In contrast, the upper reaches of all four basins, especially the Tapi basin, are mountainous terrain. Approximately 62.6% of the total area in these river basins is utilized for agriculture, and forested regions compose about 26.2%. [Base map (a) from ArcGIS World Ocean Base, Sources: Esri, Garmin, GEBCO, NOAA, NGDC, and other contributors.]

at random points, and sampling with replacement via bootstrapping. The NBRFS framework applies various nonparametric tests (e.g., paired two-sample and multisample tests) to detect significant changes in statistical properties (i.e., mean, variance, skewness, kurtosis, median, distribution, and autocorrelation) across blocks. The NBRFS framework also comprises a pool of statistical results from time-series assessments, including 17 evaluations/time series (Text S2). The use of multiple tests overcomes the limitations of individual assessments and enhances the robustness of stationarity evaluations (Teegavarapu and Sharma 2021; Singh et al. 2024).

## Flood Frequency Analysis

### Univariate Flood Frequency Analysis

The univariate FFA commenced with identifying the most appropriate statistical distribution using the Fitmethis function in MATLAB (de Castro 2025) and estimating its parameters using the maximum likelihood and Akaike information criterion (AIC) approaches (Akaike 1974), with further validation using the Kolmogorov–Smirnov (KS) test. The selected distributions included the generalized extreme value (GEV), Weibull, gamma, lognormal, logistic, and log-logistic distributions, all of which are widely used in extreme-value analysis. The GPD was specified a priori for

modeling the POT-based flood-peak series, whereas distribution fitting was carried out for all remaining series. Furthermore, median ranks of the candidate distributions were computed for each flood characteristic (i.e., peak, volume, and duration) across all catchments within a river basin to identify the most suitable common distribution. This approach enabled an objective comparison of distribution performance, leading to the selection of models that best represent extreme flood behavior and thereby improving the robustness of basin-scale flood risk assessments.

After identifying the most suitable distribution and estimating its parameters, quantiles for return periods (RPs) of 2–100 years were derived to enable comparison of the AMS and POT approaches. Uncertainty in quantile estimates was quantified using 1,000 stochastic replicates. For each return period, ranked-quantile estimates were used to obtain the 2.5th and 97.5th percentiles, defining a 95% confidence interval around the best-fit estimate. Despite neglecting other sources of uncertainty, the approach provides a useful comparative metric for evaluating performance differences between POT-based and AMS-based fitting methods (O'Shea et al. 2023).

### Bivariate Flood Frequency Analysis

The bivariate FFA first assesses the strength of association between each pair of flood characteristics using Pearson's  $r$ , Spearman's  $\rho$ ,

and Kendall's tau correlation coefficients. Pearson's  $r$  measures linear relationships but assumes normal distribution, which is unsuitable for nonlinear transformations. Spearman's rho and Kendall's tau are rank-based measures that make no distributional assumptions about bivariate data, making them more suitable for nonlinear hydrological modeling (Reddy and Ganguli 2012). Bivariate FFA, utilizing copulas (Sklar 1959), was employed to investigate the interdependence among flood characteristic pairs [i.e., flood peak and volume (P-V), volume and duration (V-D), and flood peak and duration (P-D)] (Hao and AghaKouchak 2013). Copulas are mathematical functions that join or couple two or more time-independent variables, enabling the analysis of their relationships and dependencies while preserving the characteristics of their marginal distributions (Sklar 1959; Nelsen 2006). A bivariate copula function is expressed as

$$G_{XY}(x, y) = C(G_X(x), G_Y(y)) \quad (2)$$

where,  $G_X(x)$  and  $G_Y(y)$  are the respective marginal distribution functions of the random variables  $X$  and  $Y$ , which correspond to pairs of flood characteristics (Nelsen 2006);  $G_{XY}$  is the joint distribution function of the random vector  $(X, Y)$ ; and  $C$  is the copula that links the marginal distributions.

Although event duration is recorded in integer days, copulas assume continuous margins; however, the limited discreteness is considered negligible and does not materially affect the dependence structure, and the variable therefore was retained in its original form (Khajehali et al. 2025). The potential effect of this discreteness is considered negligible and has been noted accordingly. The Multivariate Copula Analysis Toolbox (MvCAT) in MATLAB, developed by Sadegh et al. (2017), was utilized to identify the most suitable copula and estimate its parameters. This tool uses Markov chain Monte Carlo (MCMC) simulation to evaluate 26 distinct copulas in a Bayesian framework, estimating copula parameters and their uncertainties. Given their applicability in hydrological studies, this study explicitly selected four Archimedean copula functions: Frank, Clayton, Gumbel, and Joe (Tosunoglu and Singh 2018). The analysis identified optimal marginal probability distributions for each flood characteristic pair using the AIC and the Bayesian information criterion (BIC), and subsequent validation was performed using the Anderson-Darling (AD) test to ensure statistical significance at the 5% level (Schwarz 1978). The performance of these copulas was assessed using multiple goodness-of-fit metrics, including AIC, BIC, and RMS error (RMSE), to identify the most suitable copula function.

### Joint Return Period Estimation

The return period estimation is crucial in designing hydraulic structures and managing water resources. The joint return period (JRP) refers to the likelihood of an event in which one or both flood characteristics (e.g.,  $X$  and  $Y$ ) in each pair exceed their respective thresholds. It can be defined in two ways: (1) the RP when either one of the flood characteristics,  $X \geq x$  or  $Y \geq y$ , is exceeded [OR (disjunction) case]; and (2) the RP when both flood characteristics,  $X \geq x$  and  $Y \geq y$ , are exceeded [AND (conjunction) case]. This study computed JRP for the disjunction and conjunction cases based on the best-selected copula for each flood pair in a river basin. The MvCAT toolbox is limited to calculating JRPs for the disjunction (OR) scenario only. The JRPs are expressed as (Salvadori and De Michele 2004)

$$T_{OR}(x, y) = \frac{\mu}{1 - C(G_X(x), G_Y(y))} \quad (3)$$

$$T_{AND}(x, y) = \frac{\mu}{1 - G_X(x) - G_Y(y) + C(G_X(x), G_Y(y))} \quad (4)$$

where  $\mu$  = mean interarrival time between two consecutive events (i.e., equal to 1 year for the AMS, and equal to the mean number of threshold exceedances per year for the POT).

### Methodology

The methodology for this study consisted of the execution of several steps designed to analyze flood characteristics using different approaches (Fig. 2). Initially, flood characteristics (flood peak, volume, and duration) were extracted for 31 stream gauge stations, considering the AMS and POT approaches, followed by a comprehensive stationarity assessment of flood characteristics. Thereafter, univariate FFA was conducted for each series, and flood quantiles for various return periods were computed using the AMS and POT approaches. For the bivariate FFA, copulas were applied to model dependencies between pairs of flood characteristics: P-V, V-D, and P-D. Furthermore, the flood quantiles were computed for the JRPs for OR and AND cases using the best-fitting copula. Finally, the study presents the results of univariate and bivariate FFAs to provide a comprehensive understanding of complex flood dynamics for physioclimatically diverse regions.

### Results

#### Analyzing Flood Characteristics

The SFE\_IFC toolbox was used to select thresholds and their corresponding peaks per year (PPY) in the POT series and to extract flood characteristics from both the AMS and POT series (Table 1). For each station, the PPY was estimated within the toolbox as the mean annual number of threshold exceedances. Across all stations, the average PPY ranged from 1 to 5, and the threshold percentile values ranged from 87 to 99.97 percentiles (Table 1). AD test  $p$ -values exceeded 0.1 at all stations, supporting the suitability of the GPD for the POT flood-peak series, and dispersion indices close to unity and within the 95% confidence interval indicate that event counts followed a Poisson process, confirming the reliability and independence of the POT-based peak series for subsequent analysis. Building on this validation, the flood characteristics, namely volume and duration, systematically were extracted for the AMS and POT series. These characteristics served as the basis for comparing the two approaches and for providing critical insights into flood dynamics across the study region. The maximum peak flow remained identical across both approaches. However, discrepancies were observed in the maximum volume and duration for certain stream gauging stations, for which values obtained using the POT exceeded those from the AMS (Table 1). This highlights the limitation of the AMS approach in retaining crucial information for a comprehensive understanding of flood patterns. Fig. S1 depicts the spatial distribution of mean flood peak, volume, and duration across all stations determined using the AMS approach. Remarkably, higher values of all flood characteristics were observed at stations along the Narmada River than at those along other rivers. Stations with the highest flood peaks also exhibited the largest mean volumes and durations, highlighting these areas' heightened vulnerability to flood-related damage. Fig. S2 complements spatial analysis by providing temporal insights, showing that each station (i.e., S4, M4, T5, and N12) exhibits distinct temporal patterns that reflect localized hydrological responses to regional drivers.



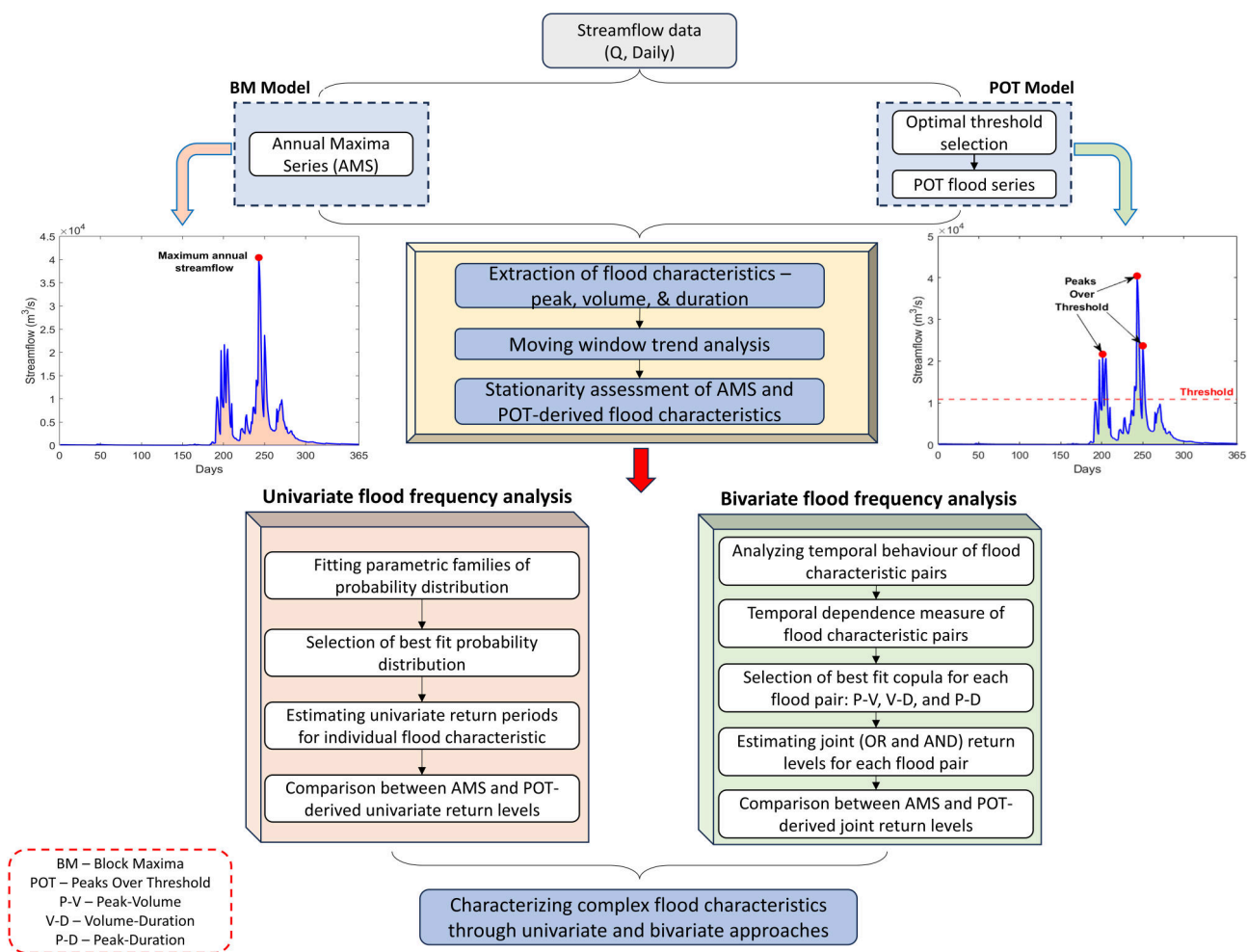


Fig. 2. Schematic representation of the methodological framework proposed in the study.

### Trend and Stationarity Assessment of Flood Characteristics

A moving-window trend analysis used the modified Mann-Kendall (MMK) test to assess trends in flood characteristics derived from the AMS and POT approaches (Fig. S3). The AMS-derived flood peak series had the highest proportion of decreasing trends, with all stations in the Narmada and Tapi basins declining [Fig. S3(a)]. In contrast, the Mahi and Sabarmati basins had mixed trends, alternating between increases and decreases, underscoring the spatial variability in hydrological responses. However, the POT-derived flood peak series exhibited mixed trends that were largely statistically insignificant across all stations, indicating greater variability or weaker trend signals than in the AMS series. The volume series from both approaches had trends like those observed in flood-peak series [Fig. S3(b)]. Both approaches indicated a higher percentage of insignificant increasing trends in the duration series [Fig. S3(c)]. Notably, variations in the threshold value used in the POT approach can influence trend magnitudes and even reverse trend direction (Agilan et al. 2021).

Trends in flood characteristics may arise from long-term persistence or limited record lengths; hence, the NBRS approach was applied to rigorously assess nonstationarity. Stationarity results for AMS [Figs. 3(a–c)] and POT [Figs. 3(d–f)] are presented and discussed in detail. The Sabarmati and Mahi basins exhibited higher NS in the POT-derived flood peak [Figs. 3(a and d)], suggesting significant temporal changes in extreme

floods. Interestingly, a greater number of stations in the Narmada basin exhibited stationary behavior in the POT-derived flood peak series, whereas they were NS in the AMS series [Figs. 3(a and d)]. The Tapi basin exhibited stationarity in the AMS and POT-derived flood-peak [Figs. 3(a and d)] and volume series [Figs. 3(b and e)], indicating time-invariance in flood behavior. For the volume series, the Narmada and Mahi River basins had more stations with NS in the AMS-derived series [Figs. 3(b and e)]. The AMS-based duration series for the Sabarmati, Tapi, and Narmada basins exhibited a higher degree of NS, suggesting temporal shifts in flood duration over time [Figs. 3(c and f)]. However, the POT approach indicated minimal NS, with only one station in each of the Sabarmati (S3), Narmada (N12), and Tapi (T5) basins showing NS behavior, indicating that the AMS method may be more sensitive to changes in flood duration. Conversely, the Mahi basin exhibited a higher number of stations with NS behavior in the POT-derived duration series [Figs. 3(c and f)]. These findings suggest that flood characteristics vary across basins, and highlight the complexities of assessing flood dynamics over time.

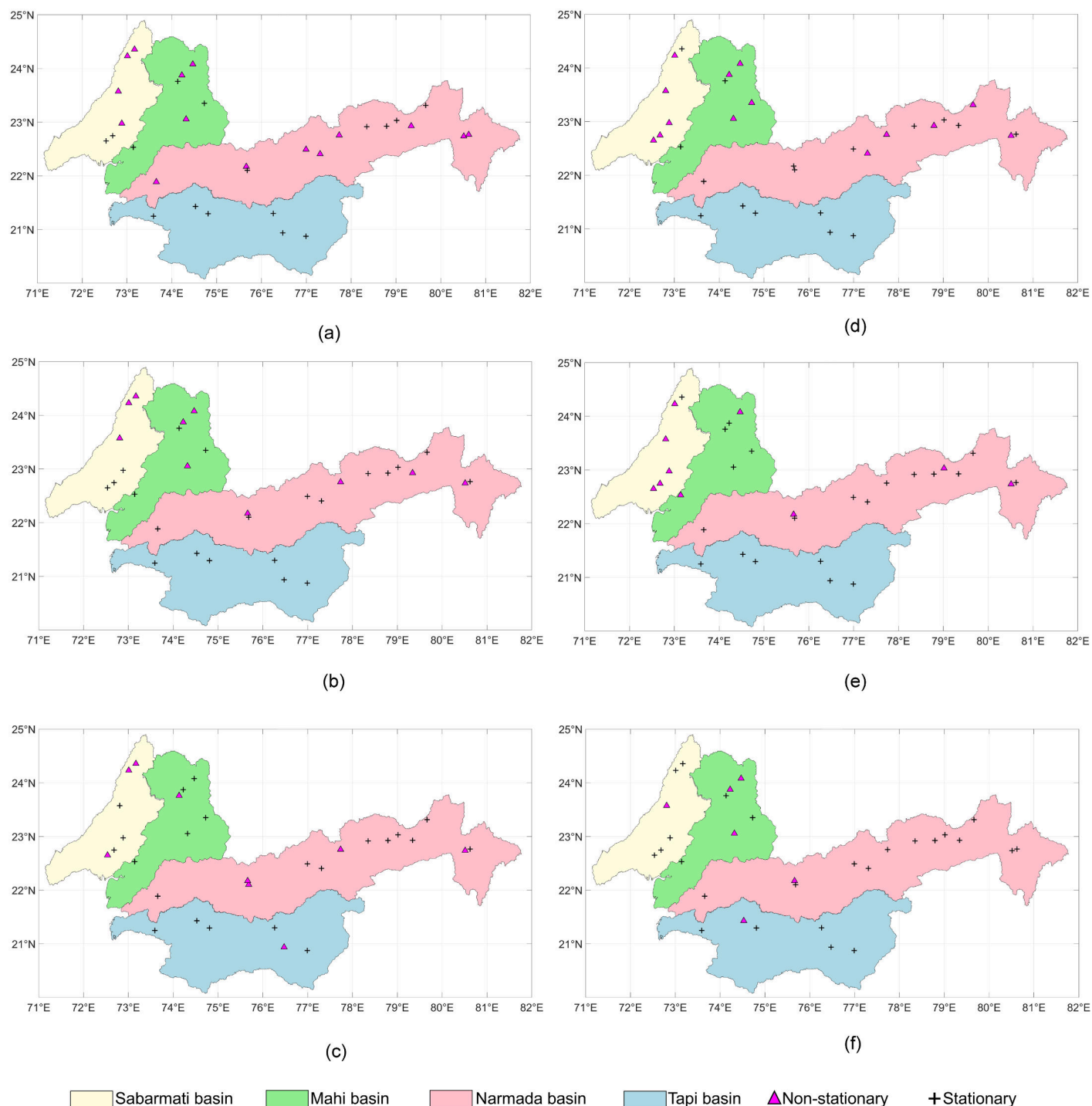
### Comparison of AMS- and POT-Derived Flood Quantiles

Univariate FFA begins by identifying the best-fit marginal distribution for each flood characteristic and each stream gauging station (Table S2). Subsequently, the best-fit distribution for the entire basin

**Table 1.** Flood characteristics derived using AMS and POT approaches for all stream gauging stations

| Notation | Stream gauging station | Data length | Annual maximum series        |            |                              |            |                     |            | Peaks-over-threshold series |                                    |                 |                              |            |                     |            |
|----------|------------------------|-------------|------------------------------|------------|------------------------------|------------|---------------------|------------|-----------------------------|------------------------------------|-----------------|------------------------------|------------|---------------------|------------|
|          |                        |             | Max peak (m <sup>3</sup> /s) | Corr. year | Max volume (m <sup>3</sup> ) | Corr. year | Max duration (days) | Corr. year | Peaks per year              | Flow threshold (m <sup>3</sup> /s) | Flow percentile | Max volume (m <sup>3</sup> ) | Corr. year | Max duration (days) | Corr. year |
| S1       | Jotasan                | 1995–2021   | 1,340                        | 2006       | 33,671                       | 2006       | 11                  | 1996       | 1.42                        | 84.1                               | 95.8            | 37,580                       | 2006       | 13                  | 2005       |
| S2       | Kheroj                 | 1992–2021   | 8,351                        | 2015       | 98,161                       | 2015       | 15                  | 2014       | 1.69                        | 231.9                              | 97.1            | 98,161                       | 2015       | 15                  | 2006       |
| S3       | Derol bridge           | 1992–2021   | 3,853                        | 2017       | 57,476                       | 2017       | 11                  | 1996       | 3.24                        | 46.4                               | 90.7            | 95,853                       | 2006       | 18                  | 2015       |
| S4       | Ratanpur               | 1991–2021   | 3,732                        | 2006       | 39,463                       | 2006       | 14                  | 2010       | 4.1                         | 63                                 | 91.0            | 39,463                       | 2006       | 18                  | 2003       |
| S5       | Kheda                  | 1989–2021   | 4,508                        | 2006       | 76,296                       | 2006       | 14                  | 2005       | 4.38                        | 48.3                               | 86.9            | 123,800                      | 2006       | 18                  | 2017       |
| S6       | Vautha                 | 2000–2021   | 7,202                        | 2005       | 202,720                      | 2005       | 14                  | 2017       | 1.48                        | 704.2                              | 96.1            | 215,360                      | 2005       | 17                  | 2003       |
| M1       | Mataji                 | 1982–2021   | 10,257                       | 1996       | 137,750                      | 1996       | 12                  | 1994       | 1.82                        | 880                                | 97.7            | 164,680                      | 2006       | 13                  | 2011       |
| M2       | Dhariawad              | 1990–2021   | 1,980                        | 2006       | 38,632                       | 2006       | 14                  | 2020       | 2.3                         | 82.9                               | 92.4            | 39,065                       | 2011       | 17                  | 2011       |
| M3       | Rangeli                | 1978–2021   | 5,179                        | 2006       | 129,120                      | 2006       | 15                  | 1992       | 3.18                        | 152.3                              | 90.4            | 129,120                      | 2006       | 19                  | 1996       |
| M4       | Paderdibadi            | 1978–2021   | 25,775                       | 2018       | 292,030                      | 2006       | 15                  | 2010       | 2.14                        | 717.5                              | 95.2            | 319,010                      | 2006       | 15                  | 2011       |
| M5       | Chakaliya              | 1991–2021   | 7,432                        | 2006       | 135,480                      | 2006       | 12                  | 2002       | 1.5                         | 1,025                              | 97.7            | 146,280                      | 2004       | 13                  | 1996       |
| M6       | Khanpur                | 1979–2021   | 31,062                       | 2006       | 663,190                      | 2006       | 14                  | 1992       | 1.55                        | 2,432                              | 97.0            | 944,520                      | 2006       | 14                  | 1991       |
| N1       | Mohgaon                | 1978–2021   | 11,600                       | 2004       | 124,780                      | 2004       | 11                  | 1979       | 4.85                        | 502.3                              | 91.6            | 136,200                      | 2004       | 16                  | 1994       |
| N2       | Manot                  | 1977–2021   | 6,806                        | 2006       | 110,490                      | 1987       | 11                  | 1993       | 3.09                        | 955.6                              | 95.5            | 130,960                      | 1994       | 15                  | 1994       |
| N3       | Patan                  | 1979–2021   | 2,202                        | 2013       | 127,000                      | 2013       | 13                  | 1980       | 3.49                        | 320.6                              | 89.7            | 131,340                      | 1994       | 18                  | 2002       |
| N4       | Belkhedi               | 1977–2021   | 7,600                        | 1994       | 98,891                       | 2018       | 12                  | 2018       | 1.23                        | 567.6                              | 98.6            | 98,891                       | 2018       | 13                  | 2019       |
| N5       | Barmanghat             | 1973–2020   | 21,500                       | 1999       | 929,130                      | 1999       | 14                  | 1999       | 1.4                         | 4,570                              | 97.0            | 929,130                      | 1999       | 16                  | 1994       |
| N6       | Gadarwara              | 1977–2021   | 10,138                       | 2016       | 104,330                      | 1999       | 11                  | 1988       | 1.67                        | 704                                | 97.7            | 187,010                      | 1999       | 14                  | 1994       |
| N7       | Sandia                 | 1978–2021   | 25,288                       | 2009       | 1,351,700                    | 1999       | 14                  | 1999       | 4.12                        | 2,150                              | 87.7            | 1,351,700                    | 1999       | 17                  | 1983       |
| N8       | Hoshangabad            | 1973–2021   | 31,600                       | 1973       | 1,816,700                    | 1999       | 14                  | 1999       | 1.43                        | 9,096                              | 97.4            | 1,826,000                    | 1999       | 15                  | 1999       |
| N9       | Chhidgaon              | 1977–2021   | 9,625                        | 2007       | 104,740                      | 1998       | 14                  | 1986       | 3.28                        | 268                                | 93.8            | 104,740                      | 1998       | 21                  | 2016       |
| N10      | Handia                 | 1977–2020   | 31,880                       | 2013       | 1,747,500                    | 1999       | 14                  | 1996       | 1.44                        | 9,192                              | 97.1            | 1,771,900                    | 1999       | 18                  | 1984       |
| N11      | Kogaon                 | 1978–2020   | 8,300                        | 1990       | 115,150                      | 2002       | 12                  | 2002       | 4.74                        | 200                                | 90.6            | 115,150                      | 2002       | 14                  | 1980       |
| N12      | Mandleshwar            | 1973–2021   | 48,200                       | 1994       | 2,158,300                    | 1999       | 15                  | 1999       | 1.35                        | 12,100                             | 97.1            | 2,158,300                    | 1999       | 15                  | 1994       |
| N13      | Garudeshwar            | 1973–2020   | 60,642                       | 1994       | 1,781,400                    | 1999       | 15                  | 1993       | 1.53                        | 10,780                             | 96.7            | 1,781,400                    | 1999       | 15                  | 1993       |
| T1       | Burhanpur              | 1973–2021   | 32,686                       | 2007       | 432,130                      | 1978       | 13                  | 1993       | 1.21                        | 3,748                              | 99.8            | 530,490                      | 1978       | 13                  | 1977       |
| T2       | Gopalkheda             | 1978–2021   | 4,124                        | 2006       | 93,135                       | 2006       | 14                  | 2017       | 1.36                        | 768.2                              | 93.6            | 93,135                       | 2006       | 13                  | 1988       |
| T3       | Yerli                  | 1973–2021   | 10,380                       | 1979       | 207,150                      | 1979       | 13                  | 2002       | 2.44                        | 824.6                              | 97.5            | 207,150                      | 1979       | 14                  | 1976       |
| T4       | Gidhade                | 1973–2016   | 26,385                       | 1979       | 797,490                      | 2006       | 13                  | 1977       | 1.84                        | 3,957                              | 99.9            | 797,490                      | 2006       | 15                  | 2002       |
| T5       | Sarangkheda            | 1978–2021   | 23,044                       | 2006       | 810,210                      | 2006       | 12                  | 1993       | 1.30                        | 4,466                              | 100.0           | 810,210                      | 2006       | 15                  | 1990       |
| T6       | Ukai dam               | 1975–2021   | 29,046                       | 2006       | 1,130,800                    | 2006       | 12                  | 1996       | 1.50                        | 4,811                              | 96.5            | 1,130,800                    | 2006       | 15                  | 2002       |

Note: Corr. year = year corresponding to maximum flood peak, volume, and duration.

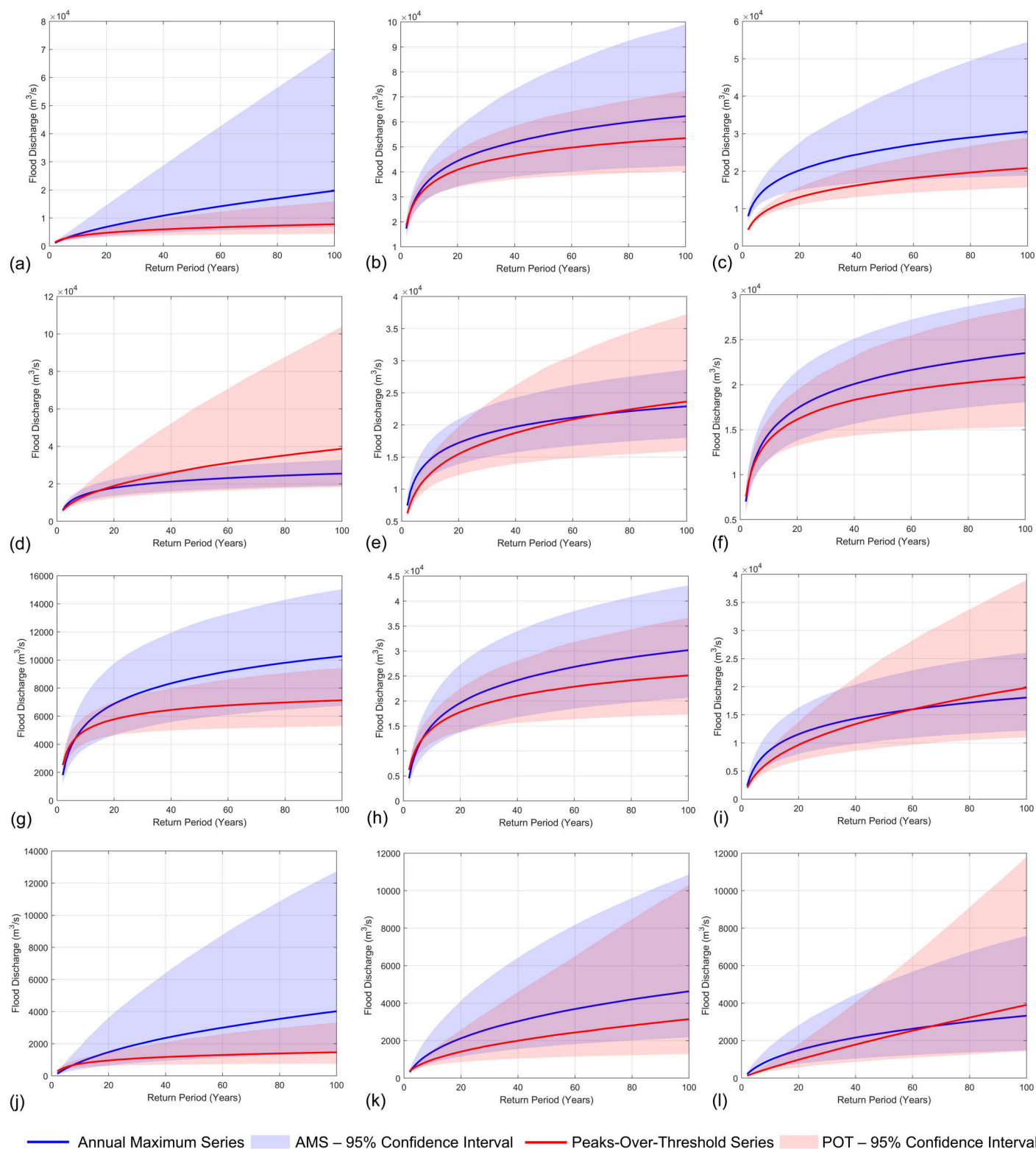


**Fig. 3.** Stationarity assessment of flood characteristics across the study domain: (a) AMS-derived peak; (b) AMS-derived volume; (c) AMS-derived duration; (d) POT-derived peak; (e) POT-derived volume; and (f) POT-derived duration.

for a particular flood characteristic was identified based on the average rank score, resulting in uniform selection across the basin. Swetapadma and Ojha (2020) observed a marginal difference in error between site-specific and basin-scale distributions, supporting the use of a standardized model across the entire basin. For the POT-derived flood peak series, the GPD was selected, because it is widely recognized as the standard for the POT approach, and its suitability was validated using the AD statistic computed via the SFE\_IFC toolbox. For the AMS-derived flood peak (volume), the best-fitting distributions were identified as GEV (GEV), gamma (lognormal),

gamma (Weibull), and lognormal (lognormal) for the Narmada, Tapi, Mahi, and Sabarmati basins, respectively. Conversely, for the volume series obtained using the POT approach, the GEV distribution was the best fit for all basins except for the Mahi basin, for which the lognormal distribution provided the best fit. Notably, the GEV distribution consistently provided the best fit for the AMS- and POT-derived duration series.

Three representative stations from each basin were selected to visually illustrate the distinct FFA patterns for the AMS- and POT-derived flood-peak quantiles (Fig. 4). After thorough analysis,



**Fig. 4.** Comparison of 2–100-year flood quantiles estimated using the AMS and POT approaches across different river basins: (a) Belkheri; (b) Garudeshwar; (c) Sandia; (d) Burhanpur; (e) Gidhade; (f) Sarangkhedda; (g) Chakaliya; (h) Khanpur; (i) Paderdibadi; (j) Jotasan; (k) Kheroj; and (l) Ratanpur. Panels (a–c) represent stations from the Narmada basin, panels (d–f) represent stations from the Tapi basin, panels (g–i) represent stations from the Mahi basin, and panels (j–l) represent stations from the Sabarmati basin.

three distinct patterns were evident when comparing flood quantiles derived from both approaches. In the first category, beyond lower return periods (i.e.,  $T > 5 - 10$  years), AMS flood quantiles consistently exceed those obtained from the POT (Fig. 4). Of the 31

stations, 25 stations (primarily from Sabarmati, Mahi, and Narmada River basins) were identified in the first category. In the second category, POT-derived flood quantiles are greater than those from AMS (for example, Burhanpur in the Tapi basin) [Fig. 4(d)]. The



remaining five stations were in the third category, in which the AMS and POT flood quantile curves intersect at approximately  $T > 50$  years, indicating varying dominance across different return periods [Figs. 4(e, i, and l)]. The uncertainty analysis shows that AMS-derived confidence intervals generally are wider than those from POT, except at the Burhanpur station, suggesting that the AMS may be more sensitive to interannual variability and extreme-event selection. Figs. S4 and S5 provide additional comparative plots of flood-peak quantiles for the remaining stations. Similar analyses were performed for flood volume and duration quantiles, yielding patterns comparable to those of flood peak quantiles (Figs. S6–S11). Notably, overlapping patterns emerged in several stations, particularly for volume quantiles, indicating minimal differences between both approaches [Figs. S6(f), S7(a, j, and k), S10(i and j), and S11(b)]. However, on most occasions, the POT-derived duration quantiles were significantly higher than those obtained using the AMS approach, highlighting the tendency of the AMS method to underestimate flood durations, especially for higher return periods (Figs. S9–S11). These differences between AMS and POT methods highlight the importance of considering regional and characteristic-specific nuances when applying FFA for hydrologic design and risk assessment.

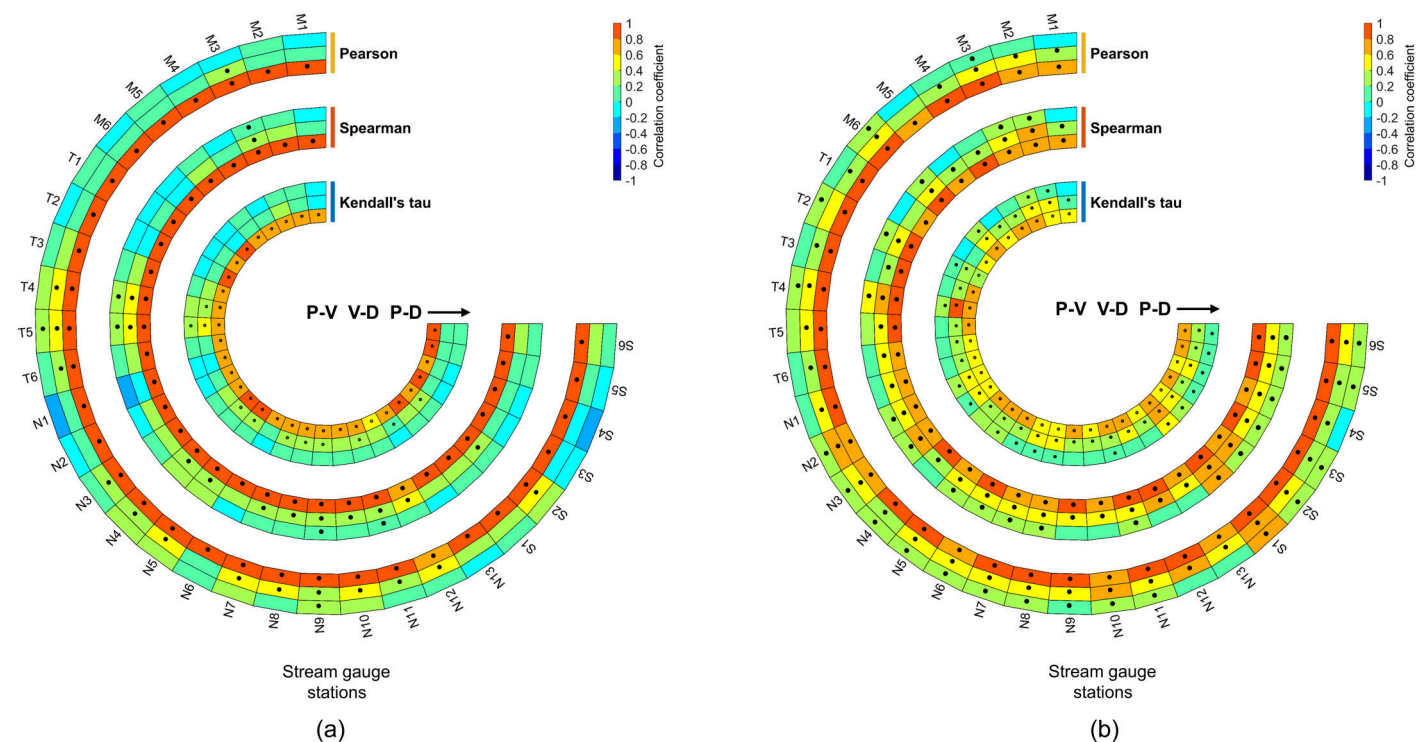
Fig. S12 illustrates the absolute difference between AMS- and POT-derived flood peak quantiles for various return periods. For smaller return periods (i.e., 2 and 5 years), flood quantiles derived from the POT approach exceeded those from the AMS approach at a significant number of stations [Figs. S12(a and b)]. However, as the return period increased to 10–50 years, all stations except Burhanpur consistently exhibited higher flood quantiles derived from the AMS approach [Figs. S12(c–f)]. Notably, Burhanpur was an exception, with POT-derived flood quantiles surpassing AMS values from the 10-year return period onward. The Sandia station (N7) had one of the highest positive deviations between AMS- and POT-derived flood quantiles for most return periods,

except  $T = 100$ . This pattern suggests that the AMS approach may overestimate flood quantiles at higher return periods because it relies on annual maxima (Fischer and Schumann 2016).

## Dependence Analysis

Understanding the dependence structure between the flood characteristics is vital for modeling their joint occurrences. The dependence between the pairs (i.e., P-V, V-D, and P-D) was analyzed for both the AMS and POT approaches (Fig. 5). Among the three correlation approaches employed, Kendall's tau consistently yielded the lowest correlation values for P-V, whereas Spearman's rho and Pearson's  $r$  correlations produced comparable results. All three methods produced closely aligned results for both P-D and V-D, demonstrating consistent relationships between these pairs across different statistical measures. Spearman correlation revealed the strongest association for P-V pairs, with values ranging from 0.77 to 0.97 (0.64–0.89) for the AMS (POT) series (Fig. 5). For the AMS series, V-D correlations were statistically significant at the 5% level for several stations in the Narmada and Tapi basins, indicating strong V-D dependence, whereas correlations in the Mahi and Sabarmati basins were largely insignificant, reflecting pronounced regional variability. However, in the POT series, V-D dependence was notably stronger and significant across nearly all stations [Fig. 5(b)]. In addition, none of the stations had negative correlations for the V-D pair using the POT approach, whereas three stations (M6, T2, and N1) in the AMS approach had negative or no correlation. This indicates that the dependence between the V-D pairs was more pronounced in the POT series than in the AMS series.

Across all basins, in the AMS approach, P-D correlations are consistently the weakest, indicating a minimal or inverse relationship between flood peak and duration at most stations [Fig. 5(a)]. This finding suggests that, in many cases, P-D



**Fig. 5.** Temporal dependence between flood characteristic pairs P-V, V-D, and P-D, analyzed using Pearson, Spearman, and Kendall's tau correlations: (a) AMS approach; and (b) POT approach. Solid circles indicate correlations that are statistically significant at a 5% significance level.

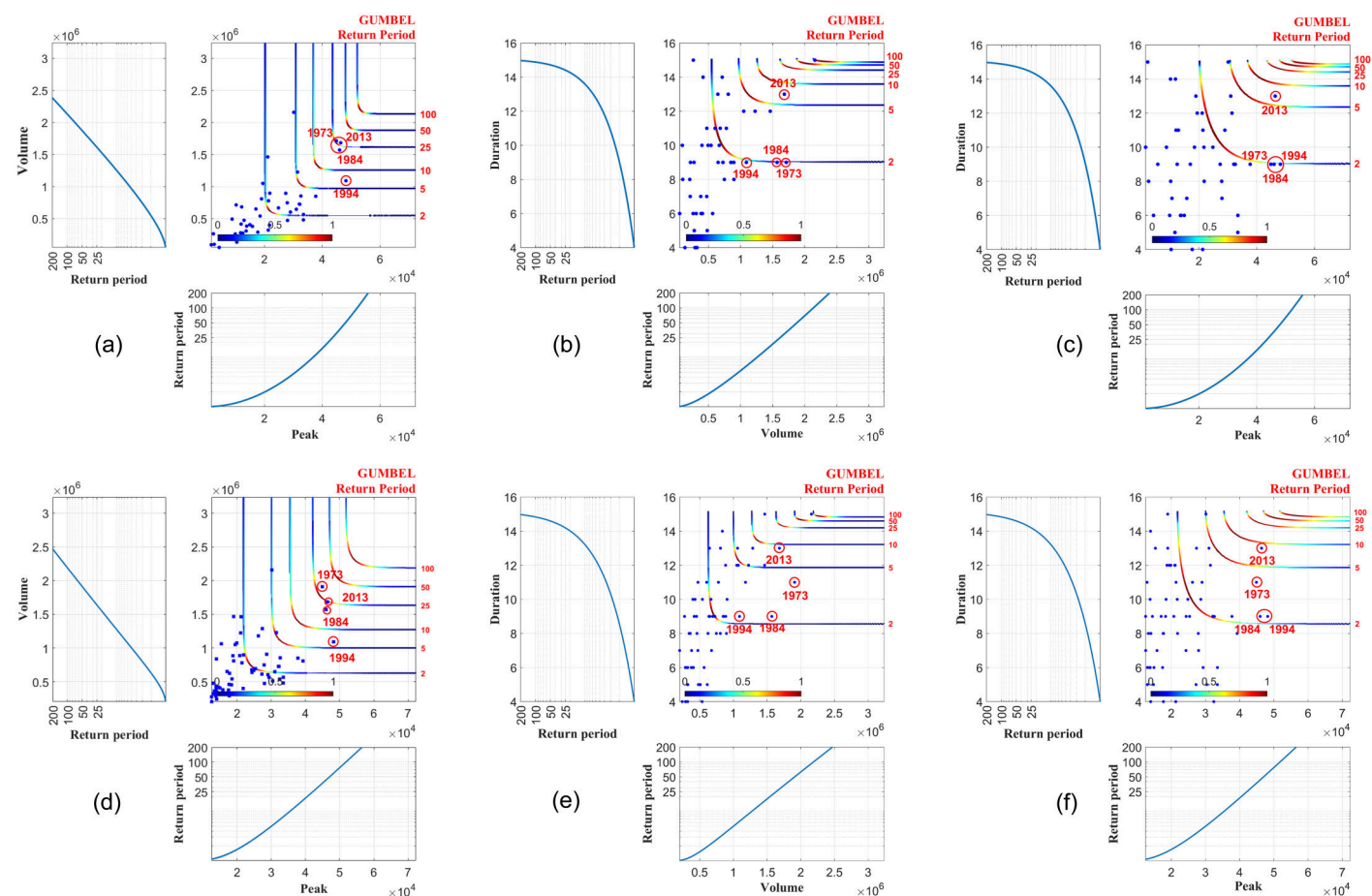
relationships are either weak or negligible. Furthermore, P-D correlations in the POT series had significance at larger stations. All stations in the Sabarmati basin exhibited consistently strong, statistically significant P-D correlations, indicating a reliable relationship between flood peak and duration in this region. Further analysis of the P-D pairs revealed that, among the 31 stations analyzed, 12 exhibited negative or minimal correlation in the AMS approach, whereas only 4 had similar behavior in the POT approach. These findings indicate that the POT approach, based on a larger sample size, provides a more reliable and consistent indication of the relationships between flood characteristic pairs. The larger data set enables a more robust characterization of the underlying dependencies among flood characteristics, yielding statistically sound insights into flood dynamics.

### Bivariate Frequency Analysis Using Copula

The correlation analysis revealed significant interdependence among flood characteristics, particularly the P-V and V-D pairs, highlighting their mutual influence. These findings are consistent with Sraj et al. (2015). The present analysis identified the best-fit copula for each flood characteristic pair at the basin scale, facilitating a uniform comparison. In this process, the copula with the highest average rank was selected as the representative copula for that basin and flood pair. Table S3 presents the ranking and RMSE values of the top three fitted

copula models for the AMS-derived P-V flood pair. Tables S4 and S5 provide the top three fitted copula models for the AMS-derived V-D and P-D flood pairs, respectively. However, only marginal differences were observed in the goodness-of-fit results among the Gumbel, Frank, and Clayton copulas, which best fit the data at most stations. The Clayton copula is best suited for modeling lower tail dependence. However, its application in this study yielded unrealistically high return periods, underestimating the risk of extreme events. Consequently, the Gumbel copula was applied to stations exhibiting a positive correlation, whereas the Frank copula was chosen as the best fit for pairs displaying negative association (Hussain et al. 2022). Table S6 summarizes the 95% uncertainty ranges associated with the selected copula parameter estimates for the AMS-derived P-V flood pair.

Given the comprehensive nature of the study, which included analyses across all stations, a representative station, N12 (i.e., Mandleshwar) in the Narmada River basin, was selected to present the copula analysis results for all three flood characteristic pairs. Fig. 6 illustrates the OR case JRP for N12, derived for each flood characteristic pair using the AMS and POT approaches. For the AMS approach, three flood events for the P-V pair occurred within the 10–25-year JRP, and no event exceeded the 25-year JRP [Fig. 6(a)]. In contrast, the POT-derived P-V series indicates that three flood events were within the 25–50-year JRP, and no event exceeded the 50-year JRP. This comparison indicates that



**Fig. 6.** Copula-based joint return periods using the Gumbel copula for the OR case for the Mandleshwar stream gauging station, Narmada River, derived for the flood characteristic pairs: (a and d) POT-derived P-V flood pairs; (b and e) POT-derived V-D flood pairs; and (c and f) POT-derived P-D flood pairs. Major flood events are marked with solid circles, and corresponding years are mentioned.

**Table 2.** Joint return periods (OR case) for flood peak and volume, derived using the AMS and POT series, at the Mandleshwar station across the Narmada River.

| Flood peak ( $\text{m}^3/\text{s}$ ) | Flood volume ( $\text{m}^3$ ) | 534,512                                                      | 930,098 | 1,239,259 | 1,693,480 | 2,083,301 | 2,521,366 |
|--------------------------------------|-------------------------------|--------------------------------------------------------------|---------|-----------|-----------|-----------|-----------|
|                                      | $T$ (years) $\rightarrow$     | 2                                                            | 5       | 10        | 25        | 50        | 100       |
|                                      | $T$ (years) $\downarrow$      | Joint return period ( $T_{OR}$ ) in years for the AMS series |         |           |           |           |           |
| 19,605                               | 2                             | 1.7                                                          | 1.9     | 2.0       | 2.0       | 2.0       | 2.0       |
| 30,392                               | 5                             | 2.0                                                          | 3.2     | 3.8       | 4.4       | 4.6       | 4.8       |
| 36,944                               | 10                            | 2.1                                                          | 3.8     | 5.3       | 7.0       | 8.1       | 8.8       |
| 44,605                               | 25                            | 2.1                                                          | 4.3     | 6.7       | 10.8      | 14.2      | 17.5      |
| 49,875                               | 50                            | 2.1                                                          | 4.5     | 7.3       | 13.0      | 19.0      | 25.9      |
| 54,782                               | 100                           | 2.1                                                          | 4.6     | 7.6       | 14.5      | 22.8      | 34.1      |
| Flood peak ( $\text{m}^3/\text{s}$ ) | Flood volume ( $\text{m}^3$ ) | 607,121                                                      | 972,467 | 128,1768  | 177,0931  | 2,221,393 | 2,759,085 |
|                                      | $T$ (years) $\rightarrow$     | 2                                                            | 5       | 10        | 25        | 50        | 100       |
|                                      | $T$ (years) $\downarrow$      | Joint return period ( $T_{OR}$ ) in years for the POT series |         |           |           |           |           |
| 607,121                              | 2                             | 2.1                                                          | 2.6     | 2.7       | 2.7       | 2.7       | 2.7       |
| 972,467                              | 5                             | 2.6                                                          | 4.8     | 6.0       | 6.6       | 6.7       | 6.8       |
| 1,281,768                            | 10                            | 2.7                                                          | 6.0     | 9.4       | 12.4      | 13.2      | 13.4      |
| 1,770,931                            | 25                            | 2.7                                                          | 6.6     | 12.4      | 23.1      | 29.4      | 32.4      |
| 2,221,393                            | 50                            | 2.7                                                          | 6.7     | 13.2      | 29.4      | 45.9      | 58.7      |
| 2,759,085                            | 100                           | 2.7                                                          | 6.8     | 13.4      | 32.4      | 58.7      | 91.7      |

the POT approach suggests a greater likelihood of severe flood events occurring at higher return periods (25–50 years) than those identified in the AMS approach [Fig. 6(d)]. This highlights the practical relevance of the POT method to capture extreme events that may not be evident in the AMS-derived series, because it accounts for larger data points, offering a more comprehensive assessment. Thus, the POT approach may provide a more accurate representation of severe and extreme flood events, especially at higher return periods. The contour line patterns of the other two pairs, V-D and P-D, differed significantly from those of the P-V pair [Figs. 6(b, c, e, and f)]. This variation can be attributed to the weaker dependence observed between these pairs. However, despite the differences in dependence strength, the results obtained for the V-D and P-D pairs were similar across the AMS and POT approaches. Notably, the return period for the same flood events was slightly higher under the POT approach than under the AMS. This suggests that, although the strength of the relationship varies between flood characteristics, the overall patterns and outcomes remain relatively consistent.

Because the P-V pair had the most substantial dependence across all basins, one representative station from each basin was selected to highlight the results for the AMS- and POT-derived P-V pair (Text S3 and Fig. S13). The analysis underscored the fundamental differences between the AMS and POT approaches in estimating JRPs, particularly for P-V pairs. The study analyzed flood events common to the AMS and POT series to compare their bivariate return period estimates. For example, at N12, the 1973 flood event exhibited a JRP of 13.9 years (42.4 years) using the AMS (POT) approach, respectively. Conversely, for the 1994 event, the JRP was 5.7 years (8.9 years) for the AMS (POT) series. This pattern underscores a key finding: for smaller JRPs, the discrepancy between the two approaches was minimal, whereas for larger flood events, the POT approach yielded considerably higher JRP values. Moreover, the consistency in results observed across stations further validates the effectiveness of the POT approach in capturing complex flood dynamics. The copula analysis results for other stations and flood characteristic pairs are detailed in Figs. S14–S43. This comprehensive analysis is valuable for understanding regional variations in flood dynamics and supports further in-depth investigations.

### Integrating Univariate and Bivariate Return Period Assessments

Building on the comparative evaluation of AMS and POT approaches in univariate and bivariate FFA, it is imperative to develop a more comprehensive framework that integrates nuanced insights from both univariate and bivariate return period assessments. This integration is a vital link in the study's overall framework, enabling the mathematical estimation of joint return periods through copula-based modeling. This approach recognizes that the univariate perspective represents a distinct statistical viewpoint compared with the multivariate framework. The univariate analysis conducted previously played a pivotal role in identifying the marginal distributions that underpin the calculation of JRPs.

The statistical results for all flood characteristic pairs at the N12 station for the OR case are presented in Tables 2–4 to further explore the flood quantiles derived from univariate and bivariate FFA. The  $T_{OR}$  values represent the JRPs (in years) estimated using the OR case of the bivariate analysis, indicating the likelihood of either flood variable (e.g., peak or volume) exceeding the specified value. The numbers above and to the left of the  $T_{OR}$  values correspond to the univariate return periods for the individual flood characteristics. JRPs for P-V show that the bivariate RPs in the AMS case generally were smaller than the univariate RPs. Conversely, for the POT case, the bivariate RPs were notably higher than the univariate RPs (Table 2). A similar trend is evident for the V-D and P-D pairs, with AMS yielding consistently smaller JRPs and POT producing comparatively larger RPs (Tables 3 and 4). For the 1972 flood event, the univariate RPs indicate that the P-V had return periods of 26 years (19.8 years) and 25.9 years (31.4 years) for the AMS (POT) series, respectively. However, the bivariate JRP for the same event was 13.9 years (42.4 years) under the AMS (POT) approach. This suggests that the POT method better captures the extremity of joint flood characteristics by accounting for more frequent threshold exceedances, whereas the AMS approach underestimates the true rarity of extreme joint occurrences.

Fig. 7 presents JRP estimates from the AMS and POT approaches for the OR and AND cases of the P-V pair across various univariate return periods, highlighting their complementary



**Table 3.** Joint return periods (OR case) for flood volume and duration, derived using the AMS and POT series, at the Mandleshwar station across the Narmada River.

| Flood volume (m <sup>3</sup> ) | Flood duration (days)     | 9                                                            | 12  | 13   | 14   | 15   | 16    |
|--------------------------------|---------------------------|--------------------------------------------------------------|-----|------|------|------|-------|
|                                | $T$ (years) $\rightarrow$ | 2                                                            | 5   | 10   | 25   | 50   | 100   |
|                                | $T$ (years) $\downarrow$  | Joint return period ( $T_{OR}$ ) in years for the AMS series |     |      |      |      |       |
| 534,512                        | 2                         | 1.5                                                          | 1.9 | 2.0  | 2.0  | 2.0  | 2.0   |
| 930,098                        | 5                         | 1.9                                                          | 3.4 | 4.3  | 4.8  | 4.9  | 5.0   |
| 1,239,259                      | 10                        | 2.0                                                          | 4.3 | 6.5  | 8.8  | 9.5  | 9.8   |
| 1,693,480                      | 25                        | 2.0                                                          | 4.8 | 8.8  | 16.0 | 20.7 | 23.3  |
| 2,083,301                      | 50                        | 2.0                                                          | 4.9 | 9.5  | 20.7 | 31.9 | 41.2  |
| 2,521,366                      | 100                       | 2.0                                                          | 5.0 | 9.8  | 23.3 | 41.2 | 63.6  |
| Flood volume (m <sup>3</sup> ) | Flood duration (days)     | 9                                                            | 11  | 13   | 14   | 15   | 15    |
|                                | $T$ (years) $\rightarrow$ | 2                                                            | 5   | 10   | 25   | 50   | 100   |
|                                | $T$ (years) $\downarrow$  | Joint return period ( $T_{OR}$ ) in years for the POT series |     |      |      |      |       |
| 607,121                        | 2                         | 2.2                                                          | 2.7 | 2.7  | 2.7  | 2.7  | 2.7   |
| 972,467                        | 5                         | 2.7                                                          | 5.3 | 6.4  | 6.7  | 6.8  | 6.8   |
| 1,281,768                      | 10                        | 2.7                                                          | 6.4 | 10.3 | 13.0 | 13.4 | 13.5  |
| 1,770,931                      | 25                        | 2.7                                                          | 6.7 | 13.0 | 25.6 | 31.6 | 33.4  |
| 2,221,393                      | 50                        | 2.7                                                          | 6.8 | 13.4 | 31.6 | 50.9 | 63.1  |
| 2,759,085                      | 100                       | 2.7                                                          | 6.8 | 13.5 | 33.4 | 63.1 | 101.7 |

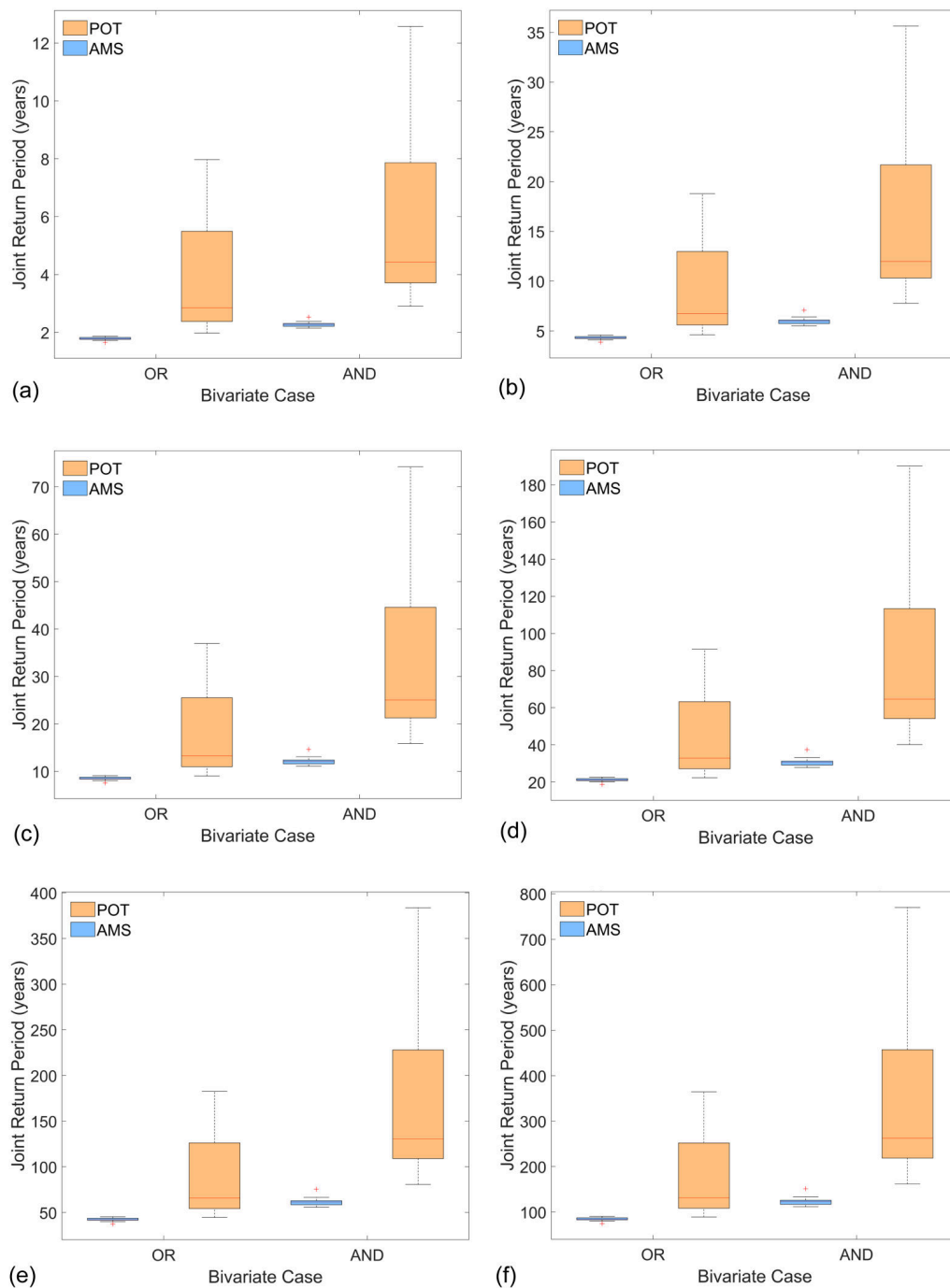
**Table 4.** Joint return periods (OR case) for flood peak and duration, derived using the AMS and POT series, at the Mandleshwar station across the Narmada River.

| Flood peak (m <sup>3</sup> /s) | Flood duration (days)     | 9   | 12                                                           | 13   | 14   | 15   | 16   |
|--------------------------------|---------------------------|-----|--------------------------------------------------------------|------|------|------|------|
|                                | $T$ (years) $\rightarrow$ | 2   | 5                                                            | 10   | 25   | 50   | 100  |
|                                | $T$ (years) $\downarrow$  |     | Joint return period ( $T_{OR}$ ) in years for the AMS series |      |      |      |      |
| 19,605                         | 2                         | 1.0 | 1.0                                                          | 1.1  | 1.1  | 1.1  | 1.1  |
| 30,392                         | 5                         | 1.2 | 1.4                                                          | 1.4  | 1.5  | 1.5  | 1.5  |
| 36,944                         | 10                        | 1.4 | 1.8                                                          | 2.0  | 2.1  | 2.2  | 2.2  |
| 44,605                         | 25                        | 1.7 | 2.7                                                          | 3.3  | 3.8  | 4.0  | 4.2  |
| 49,875                         | 50                        | 1.8 | 3.3                                                          | 4.6  | 5.9  | 6.6  | 6.9  |
| 54,782                         | 100                       | 1.9 | 3.9                                                          | 6.0  | 8.9  | 10.4 | 11.4 |
| Flood peak (m <sup>3</sup> /s) | Flood duration (days)     | 9   | 11                                                           | 13   | 14   | 15   | 15   |
|                                | $T$ (years) $\rightarrow$ | 2   | 5                                                            | 10   | 25   | 50   | 100  |
|                                | $T$ (years) $\downarrow$  |     | Joint return period ( $T_{OR}$ ) in years for the POT series |      |      |      |      |
| 9,431                          | 2                         | 1.9 | 2.3                                                          | 2.5  | 2.6  | 2.7  | 2.7  |
| 19,013                         | 5                         | 2.3 | 4.0                                                          | 5.1  | 6.1  | 6.4  | 6.6  |
| 24,566                         | 10                        | 2.5 | 5.1                                                          | 7.6  | 10.6 | 12.0 | 12.8 |
| 30,207                         | 25                        | 2.6 | 6.1                                                          | 10.6 | 18.6 | 24.5 | 28.7 |
| 33,477                         | 50                        | 2.7 | 6.4                                                          | 12.0 | 24.5 | 36.9 | 48.7 |
| 36,079                         | 100                       | 2.7 | 6.6                                                          | 12.8 | 28.7 | 48.7 | 73.4 |

perspectives on joint flood behavior. The analysis reveals that the AMS-derived JRPs exhibited comparatively lower dispersion for both bivariate cases across all univariate RPs. The AMS-based JRPs consistently were below (exceeded) the univariate return periods for the OR (AND) case. Similar behavior was observed for the POT approach, although there was a substantial difference between the AMS- and POT-derived JRPs. The POT approach had significantly greater dispersion in the JRPs, indicating greater variability in the results. To expand the analysis, the JRPs for V-D and P-D pairs also were examined across the stations (Figs. S44 and S45). The differences between AMS and POT-derived quantiles for the V-D pair diminished as the return period increased. Further analysis of the P-D pairs revealed that, in the AMS approach, the greater number of stations with high RPs resulted in a wider boxplot range, indicating greater variability (Fig. S45). Conversely, in the POT approach, stations with higher RPs were identified as outliers in

the box plot, underscoring the differences in how the two methods handle extreme values. For the OR case, the JRPs remained relatively closer to the univariate RPs across both approaches. However, in the AND case, the RPs were significantly higher, particularly in stations modeled using the Frank copula. These observations highlight the variability introduced by dependency structures and by the choice of copulas when estimating JRPs, underscoring the importance of selecting appropriate methods that reflect the data's underlying characteristics.

Building on these observations, a focused analysis was conducted on major flood events in the Narmada River basin to investigate the behavior of univariate and bivariate RPs under the AMS and POT approaches, which offers complementary insights into the joint behavior of flood characteristics (Fig. 8). Three major stations along the main river [i.e., one each in the upper Narmada (Hoshangabad), middle Narmada (Mandleshwar), and lower Narmada (Garudeshwar)] were

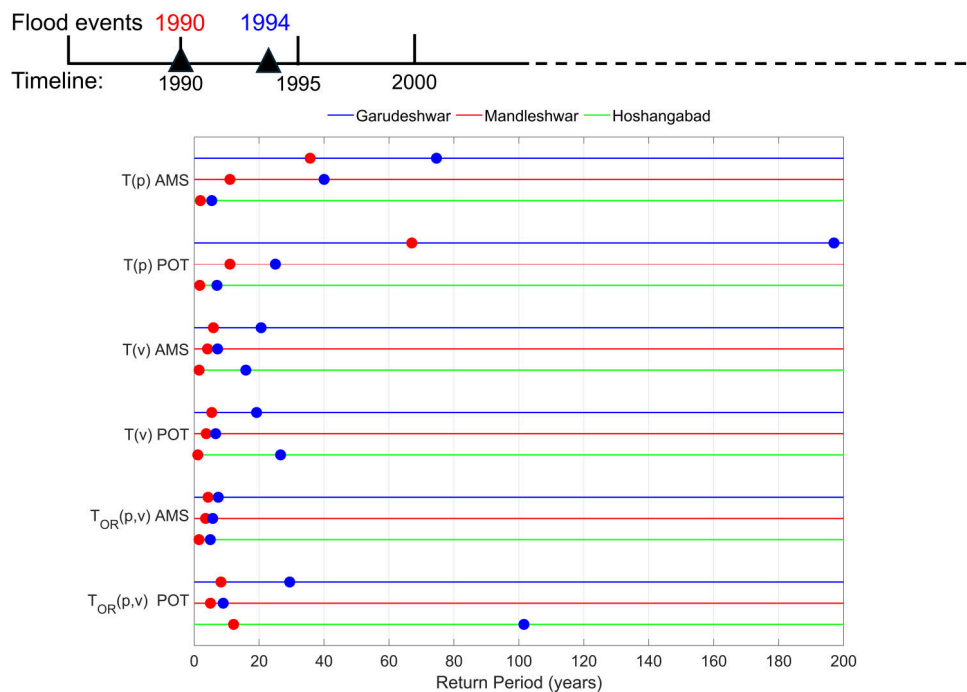


**Fig. 7.** AMS- and POT-derived univariate and bivariate (considering OR and AND cases) return periods for univariate return periods,  $T$ , of (a) 2 years; (b) 5 years; (c) 10 years; (d) 25 years; (e) 50 years; and (f) 100 years.

selected for the analysis. The analysis indicated that, in the univariate framework, the AMS and POT approaches had similar patterns, with increasing RPs from upstream to downstream, particularly for flood peaks, suggesting a higher flood risk in downstream regions. However, in the bivariate framework their behavior diverged: the AMS-based analysis tended to yield smaller JRPs, indicating a higher perceived risk, whereas the POT-based analysis generally provided higher joint RPs, reflecting its ability to capture more extreme and frequent flood events. Together, these approaches offer complementary insights into the joint behavior of flood characteristics, highlighting the distinct strengths of each method, particularly the POT approach's greater sensitivity to extreme events when multiple flood characteristics are evaluated simultaneously.

## Discussion

This study advances FFA by systematically evaluating BM and POT approaches within univariate and bivariate frameworks across four river basins with diverse physioclimatic regimes. Although simple and widely applied, the BM approach may incorporate opportunistic peaks (i.e., annual maxima from dry years that do not reflect true flood behavior), particularly in arid and semiarid systems. In contrast, the POT approach captures extreme flood processes more effectively in basins characterized by infrequent flooding or multiple flood events per year. The results show that POT-based estimates remain robust and exhibit comparatively lower uncertainty, even for short



**Fig. 8.** Representation of univariate and joint return periods, derived from the AMS and POT approaches, for the selected flood events (1990 and 1994) at the Garudeshwar, Mandleshwar, and Hoshangabad stations in the Narmada River basin. Here,  $T(p)$  and  $T(v)$  represent the univariate return period for flood peak and volume, and  $T_{OR}(p, v)$  denotes the joint return period for flood peak-volume (P-V) pair for the OR case.

record lengths, due to the increased effective sample size from threshold exceedances (Bhaktikul and Sharif 2022). These findings highlight the practical value of POT-based FFA, because its more-stable and -reliable flood quantile estimates improve hydraulic design, floodplain management, and risk-informed decision-making in data-limited regions, with gains in accuracy and reduced uncertainty outweighing the added methodological complexity (Bian et al. 2020).

Another important aspect of this study is the integration of univariate and bivariate analyses, which provide complementary insights shaped by the degree of dependence among flood variables. Although univariate methods provide a distinct statistical perspective, the bivariate approach allows a more comprehensive characterization of flood events under a changing climate and urbanization, particularly valuable in the design of dams and associated hydraulic structures, for which the joint consideration of flood magnitude, volume, and timing can improve safety and reliability. Significant correlations across most variable pairs indicate that bivariate analysis enhances understanding of flood behavior and improves flood quantile estimation, whereas weaker correlations yield comparable results, consistent with previous studies (Callau Poduje et al. 2014). Although copula parameters estimated using the MCMC method mitigate parameter uncertainty, uncertainty associated with copula selection under resampling or data perturbation has not been explicitly quantified. The framework's accuracy depends on high-quality, long-term hydrometric data, because limited coverage or short records can compromise threshold selection and copula fitting. Although it effectively models statistical dependence, it does not yet explicitly account for nonstationarity driven by land-use or climate change; future integration of nonstationary copulas and covariate-informed thresholds would further enhance its reliability. Ultimately, this improved accuracy in return periods and risk predictions empowers planners to design adaptive infrastructure and implement targeted mitigation strategies.

By integrating interacting variables, such as peak discharge, duration, and volume, this framework improves the accuracy of return period estimates and risk predictions. Such precision enables planners to design adaptive infrastructure and targeted mitigation strategies, ultimately reducing economic losses and strengthening resilience under uncertain climate conditions (Soltaninia and Eskandaripour 2025). Overall, this comprehensive approach supports informed decision-making and sustainable flood risk management across diverse physioclimatic regions.

## Conclusions

The proposed novel framework and its assessment in this study highlight that nonstationarity is increasingly prevalent in flood peaks and volumes, challenging traditional stationary assumptions in flood frequency analysis. The peaks-over-threshold approach was found to outperform the block maxima method by more effectively isolating significant flood events and reducing uncertainty. Furthermore, the results emphasize that univariate FFA is insufficient to capture the full scope of hydrological extremes; instead, a bivariate framework is essential to account for the complex dependence structures among peak, volume, and duration. By enabling flexible risk interpretation through conjunction and disjunction scenarios, this integrated framework provides a robust and scalable methodology for improving hydrologic design, early warning systems, and regional flood management.

## Data Availability Statement

The discharge data for the stream gauge stations analyzed in this study are available on the India WRIS web portal (<https://indiawris.gov.in/wris/>). The authors can provide the code used in this study upon reasonable request.



## Acknowledgments

The authors sincerely thank the India Meteorological Department, Pune; Ukai Civil Circle; India-WRIS web portal; and the Central Water Commission for providing the data necessary to conduct this study. The authors acknowledge the generous support from the JP Narayan National Centre of Excellence in the Humanities, IIT Indore (Grant No. IITI/JPNC/2023/13/P008), and the Young Faculty Research Seed Grant, IIT Indore (Grant No. IITI/YFRSG/2022-23/01), in conducting this study.

## Author Contributions

Achala Singh: Conceptualization; Data curation; Formal analysis; Methodology; Visualization; Writing – original draft. Priyank J. Sharma: Conceptualization; Funding acquisition; Methodology; Supervision; Validation; Writing – review and editing. Ramesh S. V. Teegavarapu: Methodology; Supervision; Validation; Writing – review and editing.

## Supplemental Materials

Texts S1–S3, Tables S1–S6, and Figs. S1–S45 are available in a repository online at <https://doi.org/10.5281/zenodo.19319065>.

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